

FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10 - CHEMICALS OF LIFE: FINAL EXPLANATION	
Student Assessment Document	
Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

You have finished all 6 lessons of the Chemicals of Life unit. Now write your COMPLETE EXPLANATION of the two children in Kisumu. Child A is a Standard 5 pupil who eats ugali every day and has a swollen belly, thin limbs, discoloured hair and extreme tiredness (kwashiorkor). Child B lives in the same village, eats githeri regularly and is healthy and energetic. Both children eat a similar quantity of food.

DRIVING QUESTION: Why do two children in Kisumu eating the same amount of food every day end up with completely different bodies, and what does that tell us about what food is really made of?

What to use: your notebook, your Lesson 1 prediction and first model, your food test results table, your molecular models from Lesson 4, your Driving Question Board cards and your final group poster.

Write in your own words in the space under each prompt. Use specific evidence from your lessons (what you saw in the tests and what you built), and give reasons (because...) that link each idea back to the two children. Answer in full sentences. You may add labelled diagrams. Write a focused answer to each part; about 3-4 typed pages or 4-5 handwritten pages in total is enough.

You will be marked on: understanding of the chemicals of life, use of evidence from the food tests and models, explaining the phenomenon, scientific reasoning, and applying your ideas to a real Kenyan diet.

Part 1: What Is Food Made Of?

Food is not just 'food'. (a) Name the six chemicals of life and give the main job of each in the body. (b) For each chemical, give one Kenyan food that is a good source. (c) Think about ugali and githeri. Using what you learned in Lessons 1 and 2, compare the main chemicals you would expect in each food and explain how this could matter for the two children.

Write your answer here:

Part 2: Food Tests as Evidence

Describe the food tests whose results you used in this unit. (a) For starch, protein and lipids, which test gave the evidence and what positive result is seen for each? (b) What results were recorded for ugali, for beans or githeri, and for the lipid-rich foods you tested? (c) Why is a food test better evidence than simply looking at the food?

Write your answer here:

Part 3: Structure and Function of Carbohydrates and Proteins

Use your Lesson 4 models. (a) Describe how starch is built up, starting from glucose. (b) Describe how a protein is built up, starting from amino acids, and name the bond that joins them. (c) Which element is found in amino acids but not in carbohydrates? (d) Explain why starch cannot do the jobs that protein does in the body.

Write your answer here:

Part 4: Explaining the Two Children in Kisumu

Write a Claim-Evidence-Reasoning explanation of why Child A (ugali) and Child B (githeri) have different bodies even though they eat a similar quantity of food. Your explanation must (a) make a claim, (b) use evidence from your food tests and models, and (c) explain how the chemical composition of Child A's food leads to each of these signs: swollen belly, thin limbs, discoloured hair and tiredness. (d) Look at your Lesson 3 lipid test results for ugali. What do they show, and what can they NOT show on their own?

Write your answer here:

Part 5: Applying Your Learning to a Kenyan Diet

(a) Match each condition to the chemical of life it is linked to: kwashiorkor, marasmus, scurvy, rickets, anaemia. (b) A family in the Rift Valley eats only ugali for every meal. Using what you know from your tests and models, predict which chemicals of life are plentiful and which are likely to be low, and what health problems may result. (c) Suggest two Kenyan foods you would add and explain why each helps. (d) Look back at your Lesson 1 prediction: what has changed in your thinking?

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of the chemicals of life	Correctly names all six chemicals, gives the role and a Kenyan food source for each, and accurately describes how carbohydrates and proteins are built from monomers, including the nitrogen difference.	Names most chemicals with roles and food sources; describes carbohydrate and protein structure with minor gaps or errors.	Names few chemicals or gives vague or incorrect roles; structure of carbohydrates or proteins is missing or confused.
Use of evidence from food tests and models	Accurately cites specific results (iodine blue-black in ugali, no purple Biuret in ugali, purple in beans, lipid tests negative in ugali and positive in groundnuts, avocado, coconut and milk), names the correct colour for each, and says what the tests can and cannot show.	Cites most test results and colours correctly, with some missing detail.	Evidence is general, missing or incorrect.
Explaining the phenomenon	Gives a clear Claim-Evidence-Reasoning explanation linking low protein to each sign (swollen belly, thin limbs, hair, tiredness) and explains that composition, not quantity, causes the difference.	Explains the difference with protein as the cause but links to only some signs or gives limited reasoning.	States that ugali is bad or lacks something without a chemical explanation, or focuses on quantity.
Scientific reasoning and	Logically connects evidence to	Applies ideas to a diet with reasonable	Applications are unclear or unsupported;

application	conclusions, separates what the tests showed from what is inferred about lipids and vitamins, applies ideas to a Kenyan diet with sensible food suggestions, and reflects on how thinking changed.	suggestions; reasoning mostly clear.	little reasoning or reflection.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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